

JKBOSE 2026

CLASS 11TH

**LIMITS AND
DERIVATIVES**

MATHS

LECTURE-6

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TOPICS TO BE COVERED

1

Derivatives of T-functions

2

Basic Rules of Differentiation

3

Related Problems

$$f'(a) = Rf'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\sin x - \sin y = 2 \cos\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right)$$

$$\cos x - \cos y = -2 \sin\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right)$$

* $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \rightarrow \text{exist finitely (function is differentiable)}$

$$\lim_{x \rightarrow a^+} \frac{f(x) - f(a)}{x - a} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = Rf'(a)$$

$$\lim_{x \rightarrow a^-} \frac{f(x) - f(a)}{x - a} = \lim_{h \rightarrow 0} \frac{f(a-h) - f(a)}{-h} = Lf'(a)$$

Derivative of T – Functions:

★ **Derivative of $\sin x$** : Suppose that $f(x) = \sin x$ be the given function. Then by first principle (**ab-initio method**), we have;

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{2 \cos\left(\frac{x+h+x}{2}\right) \sin\left(\frac{x+h-x}{2}\right)}{h}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{2 \cos\left(\frac{2x+h}{2}\right) \sin\left(\frac{h}{2}\right)}{h}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \left[\cos\left(\frac{2x+h}{2}\right) \cdot \frac{\sin\left(\frac{h}{2}\right)}{\left(\frac{h}{2}\right)} \right]$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \cos\left(\frac{2x+h}{2}\right) \cdot \lim_{\frac{h}{2} \rightarrow 0} \frac{\sin\left(\frac{h}{2}\right)}{\left(\frac{h}{2}\right)}$$

$$\Rightarrow f'(x) = \cos\left(\frac{2x+0}{2}\right) \times 1 \quad \left(\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \right)$$

$$\Rightarrow f'(x) = \cos\left(\frac{2x}{2}\right)$$

$$\Rightarrow f'(x) = \cos x \text{ or } \frac{d}{dx}(\sin x) = \cos x.$$

★ **Derivative of $\cos x$** : Suppose that $f(x) = \cos x$ be the given function. Then by first principle (ab-initio method), we have;

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{-2 \sin\left(\frac{x+h+x}{2}\right) \sin\left(\frac{x+h-x}{2}\right)}{h}$$

$$\Rightarrow f'(x) = - \lim_{h \rightarrow 0} \frac{2 \sin\left(\frac{2x+h}{2}\right) \sin\left(\frac{h}{2}\right)}{h}$$

$$\Rightarrow f'(x) = - \lim_{h \rightarrow 0} \left[\sin\left(\frac{2x+h}{2}\right) \cdot \frac{\sin\left(\frac{h}{2}\right)}{\left(\frac{h}{2}\right)} \right]$$

$$\Rightarrow f'(x) = - \lim_{h \rightarrow 0} \sin\left(\frac{2x+h}{2}\right) \cdot \lim_{h \rightarrow 0} \frac{\sin\left(\frac{h}{2}\right)}{\left(\frac{h}{2}\right)}$$

$$\Rightarrow f'(x) = - \sin\left(\frac{2x+0}{2}\right) \cdot 1 \left(\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \right)$$

$$\Rightarrow f'(x) = - \sin\left(\frac{2x}{2}\right) = - \sin x$$

$$\Rightarrow f'(x) = -\sin x \text{ or } \frac{d(\cos x)}{dx} = -\sin x$$

★ **Derivative of $\tan x$** : Suppose that $f(x) = \tan x$ be the given function. Then by first principle (**ab-initio method**), we have;

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\tan(x+h) - \tan x}{h}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{1}{h} \left\{ \frac{\sin(x+h)}{\cos(x+h)} - \frac{\sin x}{\cos x} \right\}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{1}{h} \left\{ \frac{\sin(x+h)\cos x - \cos(x+h)\sin x}{\cos(x+h)\cos x} \right\}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{\sin(x+h-x)}{\cos(x+h)\cos x} \right)$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{\sin h}{h \cos(x+h)\cos x}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \left\{ \frac{\sin h}{h} \times \frac{1}{\cos(x+h)\cos x} \right\}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{\sin h}{h} \times \lim_{h \rightarrow 0} \frac{1}{\cos(x+h)\cos x}$$

$$\Rightarrow f'(x) = 1 \times \frac{1}{\cos(x+0)\cos x} = \frac{1}{\cos x \cdot \cos x}$$

$$\Rightarrow f'(x) = \frac{1}{\cos^2 x} = \sec^2 x.$$

$$\Rightarrow f'(x) = \sec^2 x \text{ or } \frac{d}{dx}(\tan x) = \sec^2 x.$$

Hence the result.

★ **Derivative of $\sec x$** : Suppose that $f(x) = \sec x$ be the given function. Then by first-principle (**ab-initio method**), we have;

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\sec(x+h) - \sec x}{h}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{1}{h} \left\{ \frac{1}{\cos(x+h)} - \frac{1}{\cos x} \right\} = \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{\cos x - \cos(x+h)}{\cos(x+h) \cos x} \right)$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{1 - 2 \sin\left(\frac{x+x+h}{2}\right) \sin\left(\frac{x-x-h}{2}\right)}{h \cos(x+h) \cos x}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{1 - 2 \sin\left(\frac{2x+h}{2}\right) \sin\left(-\frac{h}{2}\right)}{h \cos(x+h) \cos x}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{2 \sin\left(\frac{2x+h}{2}\right) \sin\left(\frac{h}{2}\right)}{h \cos(x+h) \cos x}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \left\{ \frac{\sin\left(\frac{2x+h}{2}\right)}{\cos(x+h) \cos x} \times \frac{\sin\left(\frac{h}{2}\right)}{\left(\frac{h}{2}\right)} \right\}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{\sin\left(\frac{2x+h}{2}\right)}{\cos(x+h) \cos x} \times \lim_{\frac{h}{2} \rightarrow 0} \frac{\sin\left(\frac{h}{2}\right)}{\frac{h}{2}}$$

$$\Rightarrow f'(x) = \frac{\sin\left(\frac{2x+0}{2}\right)}{\cos(x+0) \cos x} \times 1 \left(\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \right)$$

$$\Rightarrow f'(x) = \frac{\sin(\frac{2x}{2})}{\cos x \times \cos x} = \frac{\sin x}{\cos x} \times \frac{1}{\cos x}$$

$$\Rightarrow f'(x) = \sec x \tan x \text{ or } \frac{d}{dx}(\sec x) = \sec x \tan x$$

★ Derivative of $\cot x$ (Homework)

★ Derivative of $\operatorname{cosec} x$ (Homework)

Example Find the derivative of e^x by first principle.

(Sol) Suppose that $f(x) = e^x$ be the given function.
Then by first principle (ab-initio method), we have;

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \left(\frac{e^{x+h} - e^x}{h} \right)$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} e^x \left(\frac{e^h - 1}{h} \right) = e^x \lim_{h \rightarrow 0} \left(\frac{e^h - 1}{h} \right)$$

$$\Rightarrow f'(x) = e^x \times 1 \left(\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1 \right)$$

$$\Rightarrow f'(x) = e^x \text{ or } \frac{d}{dx}(e^x) = e^x$$

Basic Rules of Differentiation:

Suppose that U and V be any functions of x . Also, let

C be any Constant. Then;

$$(1) \frac{d}{dx}(C) = 0$$

$$(2) \frac{d}{dx}(U+V) = \frac{dU}{dx} + \frac{dV}{dx}$$

$$(3) \frac{d}{dx}(U-V) = \frac{dU}{dx} - \frac{dV}{dx}$$

$$(4) \frac{d}{dx}(CU) = C \frac{dU}{dx}$$

$$(5) \frac{d}{dx}(UV) = U \frac{dV}{dx} + V \frac{dU}{dx} \text{ (Product rule)}$$

$$(6) \frac{d}{dx}\left(\frac{U}{V}\right) = \frac{V \frac{dU}{dx} - U \frac{dV}{dx}}{V^2} \text{ (Quotient rule)}$$

Example: find the derivative of $f(x) = e^x \cos x$.

(Sol) Given that $f(x) = e^x \cos x$

$$\Rightarrow f'(x) = \frac{d}{dx}(e^x \cos x) = e^x \frac{d}{dx}(\cos x) + \cos x \frac{d}{dx}(e^x)$$

$$\Rightarrow f'(x) = e^x \cos x - e^x \sin x = e^x (\cos x - \sin x)$$

$$\Rightarrow f'(x) = e^x (\cos x - \sin x)$$

EXAMPLE

Find the derivative of $\frac{\cos x}{1 + \sin x}$.

(Sol) Suppose that $f(x) = \frac{\cos x}{1 + \sin x}$ be the given function. Then

$$f'(x) = \frac{d}{dx} \left(\frac{\cos x}{1 + \sin x} \right) = \frac{(1 + \sin x) \frac{d(\cos x)}{dx} - \cos x \frac{d(1 + \sin x)}{dx}}{(1 + \sin x)^2} \quad (\text{Quotient rule})$$

$$\Rightarrow f'(x) = \frac{-(1 + \sin x) \sin x - \cos x(0 + \cos x)}{(1 + \sin x)^2}$$

$$\Rightarrow f'(x) = \frac{-\sin x - \sin^2 x - \cos^2 x}{(1 + \sin x)^2}$$

$$\Rightarrow f'(x) = \frac{-(\sin x + \sin^2 x + \cos^2 x)}{(1 + \sin x)^2}$$

$$\Rightarrow f'(x) = \frac{-(\cancel{\sin x} + 1)}{(\cancel{\sin x} + 1)^2}$$

$$\Rightarrow f'(x) = -\frac{1}{(1 + \sin x)}$$

Hence the result.

QUESTION

The derivative of $\left(x + \frac{1}{x}\right)^3$ is :

$$\frac{d}{dx} x^n = nx^{n-1}$$

$$(\bar{x}^3)' = -3\bar{x}^{3-1} = -3\bar{x}^{-4} =$$

1 $3x^2 + \frac{3}{x^4} + 3 + \frac{3}{x^2}$

2 $3x^2 - \frac{3}{x^4} + 3 - \frac{3}{x^2}$

3 $3x^2 + \frac{3}{x^4} + 3 - \frac{3}{x^2}$

4 None of these

(Sol) let $f(x) = \left(x + \frac{1}{x}\right)^3 = x^3 + \frac{1}{x^3} + 3x \cdot \frac{1}{x} \left(x + \frac{1}{x}\right)$

$$\Rightarrow f(x) = x^3 + \bar{x}^3 + 3\left(x + \frac{1}{x}\right)$$

$$\Rightarrow f(x) = x^3 + \bar{x}^3 + 3x + 3\bar{x}$$

$$\Rightarrow f'(x) = 3x^2 - 3\bar{x}^4 + 3 - 3\bar{x}^2$$

$$\Rightarrow f'(x) = 3x^2 - \frac{3}{x^4} + 3 - \frac{3}{x^2}$$

QUESTION

The derivative of the function $f(x) = (3x + 5)(1 + \tan x)$

1 ☒ $3x\sec^2 x - 6\sec^2 x - 3 + 3\tan x$

2 ☒ $3x\sec^2 x - 5\sec^2 x - 3 + 3\tan x$

3 ☒ $3x\sec^2 x + \sec^2 x - 3 + 3\tan x$

4 ☒ None of the above

(Sol) Given that $f(x) = (3x + 5)(1 + \tan x)$
 $\Rightarrow f'(x) = (3x + 5) \frac{d}{dx}(1 + \tan x) + (1 + \tan x) \frac{d}{dx}(3x + 5)$
 $\Rightarrow f'(x) = (3x + 5) \sec^2 x + (1 + \tan x)(3)$
 $\Rightarrow f'(x) = 3x\sec^2 x + 5\sec^2 x + 3 + 3\tan x$

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Thank You